Short answer Questions

1. Write down the possible isomers of the $C_4H_9NO_2$ give their IUPAC names
2. There are two isomers with the formula CH_3NO_2. How will you distinguish between them?
3. What happens when
   i. 2 – Nitropropane boiled with HCl
   ii. Nitrobenzene undergo electrolytic-reduction in strongly acidic medium.
   iii. Oxidation of tert – butylamine with $KMnO_4$
   iv. Oxidation of acetoneoxime with trifluoroperoxy acetic acid.
4. How will you convert nitrobenzene into
   i. 1,3,5 - trinitrobenzene
   ii. o and p- nitrophenol
   iii. m – nitro aniline
   iv. azoxybenzene
   v. hydrozobenzene
   vi. N – phenylhydroxylamine
   vii. aniline
5. Identify compounds A,B and C in the following sequence of reactions.

i) $C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273K]{HNO_2} B \xrightarrow{C_6H_5OH} C$

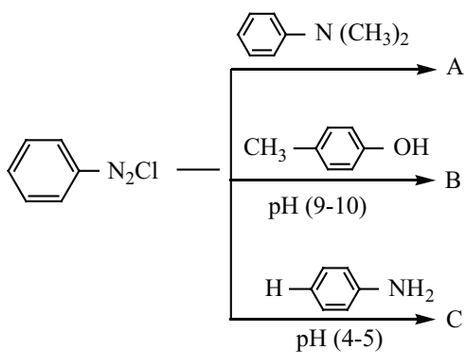
ii) $C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O\,/\,H^+} B \xrightarrow{NH_3} C$

iii) $CH_3CH_2I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow{NaOH + Br_2} C$

iv) $CH_3NH_2 \xrightarrow{CH_3\ Br} A \xrightarrow{CH_3COCl} B \xrightarrow{B_2H_6} C$

v) $C_6H_5NH_2 \xrightarrow[\text{pyridine}]{(CH_3CO)_2O} A \xrightarrow[H_2SO_4,288K]{HNO_3} B \xrightarrow{H_2O\,/\,H^+} C$

vi)



vii) $CH_3CH_2NC \xrightarrow{HgO} A \xrightarrow{H_2O} B \xrightarrow[ii)H_2O]{i)NaNO_2\ /\ HCl} C$

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